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Article

Keywords:

Posted Date: October 1st, 2024

DOI: https://doi.org/10.21203/rs.3.rs-4610184/v1

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Additional Declarations: There is **NO** Competing Interest.

Version of Record: A version of this preprint was published at Nature Computational Science on May 28th, 2025. See the published version at https://doi.org/10.1038/s43588-025-00796-8.

A computational approach to evaluate how molecular mechanisms impact large-scale brain activity

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Abstract

Assessing the impact of pharmaceutical compounds on brain activity is a critical issue in contemporary neuroscience. Currently, no systematic approach exists for evaluating these effects in whole-brain models, which typically focus on macroscopic phenomena, while pharmaceutical interventions operate at the molecular scale. Here, we address this issue by presenting a computational approach for brain simulations using biophysically grounded mean-field models that integrate membrane conductances and synaptic receptors, showcased in the example of anesthesia. We show that anesthetics targeting $GABA_A$ and NMDA receptors can switch brain activity to generalized slow-wave patterns, as observed experimentally in deep anesthesia. To validate our models, we demonstrate that these slow-wave states exhibit reduced responsiveness to external stimuli and functional connectivity constrained by anatomical connectivity, mirroring experimental findings in anesthetized states across species. Our approach, founded on mean-field models that incorporate molecular realism, provides a robust framework for understanding how molecular-level drug actions impact whole-brain dynamics.

1 Introduction

Recent advances in neuroimaging techniques have made it possible to build detailed brain atlases and generate personalized connectomes of the human brain through cutting-edge tools such as diffusion MRI, MEG, and fMRI. These advancements have catalyzed the conceptualization within the scientific community of highly detailed, personalized whole-brain models, colloquially termed "Digital Twins," poised to revolutionize the exploration, diagnosis, and therapeutic interventions for various neurological conditions such as disorders of consciousness, epilepsy, Alzheimer's disease, and schizophrenia (Arbabyazd et al., 2021; Stefanovski et al., 2021; Jirsa et al., 2023; Luppi et al., 2023; Amunts et al., 2024).

Despite significant advances in this domain, a primary challenge faced by these models lies in effectively addressing the diverse spatial and temporal scales inherent to brain activity. Numerous diseases or alterations in brain states emerge at the molecular or cellular level, requiring comprehensive descriptions within whole-brain models to ensure accuracy. Thus, integrating cellular-scale details is imperative for these models to capture the intricacies of neurological phenomena. However, the human brain comprises approximately 85 billion neurons and an estimated 10^{15} synaptic connections. Consequently, creating a large-scale model of the brain (macro-scale) with single-cell resolution (micro-scale) remains an insurmountable challenge. Establishing a seamless interface between these disparate scales is, therefore, a pivotal challenge for enhancing the utility and precision of digital twins and whole-brain simulations.

This article introduces a novel framework aimed at addressing this challenge, offering an efficient pathway for transitioning from the cellular scale to the whole-brain level. Central to our methodology is the integration of a biophysically grounded mean-field model, encompassing cellular and molecular properties. This model effectively reduces system dimensionality while capturing emergent properties observed at the meso and macro scale due to cellular-level changes. Our methodology comprises four parts: i) Single-cell models (microscale); ii) Networks of spiking neurons (mesoscale); iii) a mean-field model elucidating the activity and dynamics of the network (mesoscale, low-dimensional); and iv) a Whole-brain model leveraging the mean-field model for generating large-scale simulations (macroscale). To facilitate the execution of these simulations, we utilize The Virtual Brain (TVB) platform (Sanz-Leon et al., 2015), a versatile tool enabling the execution of large-scale simulations with customizable brain connectomes.

To illustrate our approach, we simulate the molecular actions leading to general anesthesia, specifically with anesthetics such as ketamine and propofol, and investigate their effects at the mesoscale and whole-brain levels. Our focus includes the generation of slow-wave activity, reduced responsiveness to external stimuli, and increased anatomical-functional correlations, as measured experimentally for anes-

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thetized states.

2 Results

The components of our method are introduced sequentially, following the bottom-up approach of the model construction. As a benchmark, we apply this model within the context of general anesthesia, which serves as an example of how molecular changes at the synaptic level can significantly impact whole-brain dynamics. The diagram of the framework is presented in Fig. 1.

2.1 Single-Cell Model

The first step of our framework consists of selecting a spiking neuron model. Here, we used the Adaptive Exponential integrate-and-fire model (AdEx), recognized for its capacity to capture a wide range of neuronal firing patterns, while also providing biophysically realistic representations of neuronal behavior and mathematical simplicity (Brette and Gerstner, 2005; Naud et al., 2008). Nevertheless, our methodology remains adaptable to a variety of neuronal models, such as quadratic Integrate-and-Fire or Hodgkin-Huxley (Carlu et al., 2020; Alexandersen et al., 2024; Lorenzi et al., 2023), thereby showcasing the broad applicability of our approach.

The AdEx model consists of two differential equations. The first equation (Eq. 1) describes the membrane potential dynamics and the second (Eq. 2), a slow adaptation current (see Methods section for details). This adaptation current varies over time in response to spiking activity, modulating the excitability of the neuron (Brette and Gerstner, 2005).

We consider two types of cells: pyramidal (excitatory) neurons and fast-spiking (inhibitory) interneurons. Each type of cell is characterized by a different set of parameters in the AdEx model. The baseline parameters used for each neuron type are extracted from fits on experimental data of cortical neurons (Di Volo et al., 2019). However, the method we present can be equally applied to simulate the dynamics of neurons in other brain areas such as the thalamus (Overwiening et al., 2023), basal ganglia (Tesler et al., 2023a), or cerebellum (Lorenzi et al., 2023).

2.2 Network of Spiking Neurons

The second step of our framework consists of the construction of networks of spiking neurons representing local circuits or specific brain structures. The networks are characterized by the type and number of neurons, their connectivity patterns (probability of connections between neurons), and the synaptic model used. In terms of the number of neurons and the connectivity pattern, the example presented in this article adopts values representative of prototypical cortical circuits, specifically modeling a single cortical column. We consider a network made of 10000 neurons, with 80% of pyramidal cells and 20% of interneurons, connected randomly with probability $p = 0.05$. Different connectivity patterns can be considered and are compatible with our method, which can be used to represent different brain structures (Tesler et al., 2022; Lorenzi et al., 2023; Overwiening et al., 2023; Tesler et al., 2024). Regarding the synaptic model, we consider conductance-based synapses through glutamate (AMPA-NMDA) and $GABA_A$ receptors, but current-based synapses are also compatible with our framework (El Boustani and Destexhe, 2009). The equation for the synaptic model is presented in the Methods section, Eq. 4.

2.3 Mean-Field Model

The central component of our framework is the integration of a recently developed mean-field formalism, which enables an efficient transition between scales while preserving the characteristics and relevant parameters of both the cellular and network levels. The derivation of the mean-field model used here follows a bottom-up approach based on a second-order Master Equation formalism, that describes the average activity (mean firing rate, membrane potential, conductance) and covariance of networks of spiking neurons (El Boustani and Destexhe, 2009). A key component of these equations is the transfer function, which relates a neuron's output (firing rate) to its presynaptic input. This function cannot be expressed analytically for complex neurons, particularly those interacting through conductance-based synapses. However, a semi-analytic approach can be followed. First, numerical simulations of the desired neuron model are realized and used to fit an analytically derived mathematical template of the transfer function. This approach has been applied previously to the AdEx model (Zerlaut et al., 2018; Di Volo et al., 2019), but it has been extended to more complex models, such as the quadratic integrate-and-fire neuron (Alexandersen et al., 2024), the Morris-Lecar and Hodgkin-Huxley model (Carlu et al., 2020)

Through this semi-analytical approach, population models that preserve biological relevance can be constructed. This novel formulation allows us to study the effects of changes in molecular and network parameters at the mesoscale level, trace the mechanisms behind dynamical transitions in large populations (facilitated by the low dimension of the mean-field model), and to construct finally whole-brain simu-

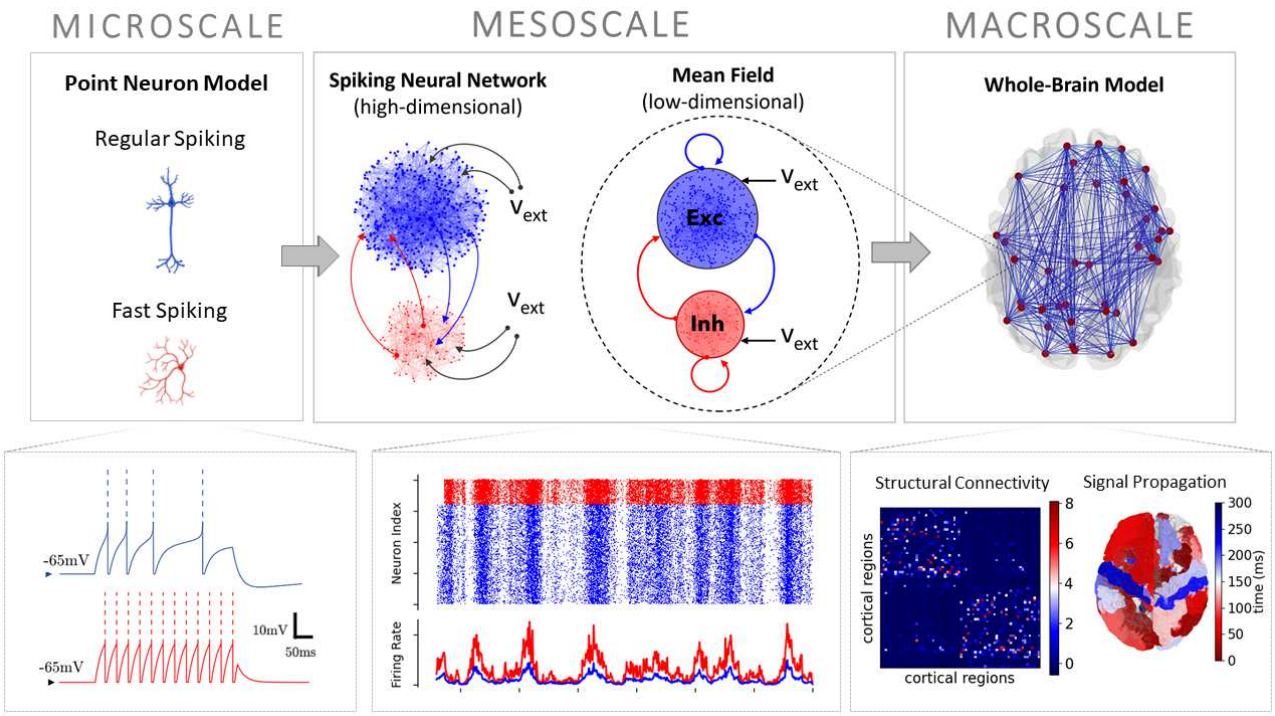

Figure 1: Schematic diagram of the bottom-up construction of the whole-brain modelling framework (top row) and the corresponding signals simulated across the various scales (bottom row). Microscale: a point neuron model is selected to describe the activity of individual neurons, representing different cellular types with specific neuroreceptors and characterized by distinct firing patterns—ranging from regular spiking cells with adaptive firing rates to fast-spiking cells. Mesoscale: based on experimental data of brain microcircuits, we connect the single neurons to build spiking neural networks (high-dimensional mesoscale) from which biologically grounded mean-field models are derived to describe collective dynamics (low- dimensional mesoscale). Macroscale: structural connectivity data are integrated into the framework to build large-scale networks of mean-field models to perform whole-brain simulations, allowing the exploration of emergent properties at this scale (e.g. responsiveness to external stimulation).

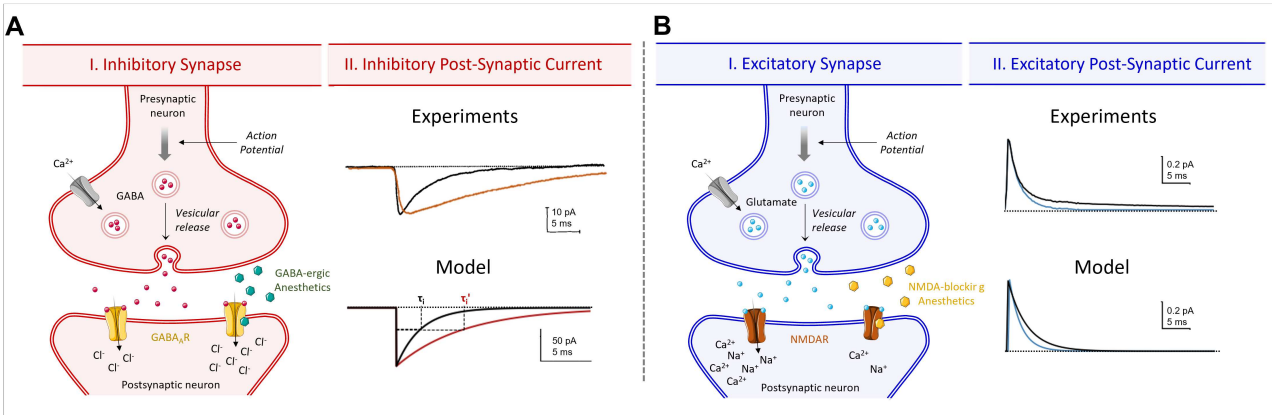

Figure 2: Synaptic targets of general anesthesia and its effect at the cellular scale, in experiments and modeling. **A.** General anesthesia that targets $GABA_A$ receptors (e.g., propofol) prolongs inhibitory post-synaptic potentials (IPSPs) by increasing the opening duration of the chloride channels. We model this effect as an increase in the inhibitory synaptic decay time constant (τ_i). We illustrate the comparison of the model with whole-cell patch clamp recordings demonstrating the prolongation of IPSP caused by halothane (top row in A.II, modified from Maciver et al. (1991). **B.** NMDA blockers (e.g. ketamine, xenon) decrease the duration of excitatory post-synaptic potentials (EPSPs), as shown by patch-clamp recordings of AMPA and NMDA currents under the effect of ketamine (top row in B.II modified from Destexhe et al. (1998)). This effect can be modeled as a decrease of the excitatory synaptic decay time constant (τ_e).

lations that incorporate the details and specificities of the micro- and mesoscale at a low computational cost.

2.4 Whole-brain model

We move to the last scale of our formalism, i.e. the whole brain scale, which is essential for examining properties that manifest exclusively at a macroscale. The whole-brain model is described by a network of mean-fields interconnected based on experimental data (connectome). First, a parcellation of the brain is selected, that divides the cortex into distinct regions. The granularity of the parcellation can vary from tens to thousands of regions. Information on the connection between these regions is available through various anatomical connectivity studies, obtained using diffusion weight imaging, tracer studies (in non-human species), or a combination of both. The output of these studies relevant to our method is a connectivity matrix that provides information on the connection strength and the distance between each pair of regions.

The activity of each region can be described using the developed mean-field model, forming a node in the whole-brain model. Each node's input consists of an external noise, described by an Ornstein-Uhlenbeck process, and the output firing rate of its connected nodes. The input of the connected nodes is weighted according to the connection strength and received with a delay based on the distance between the two nodes.

We employed a parcellation scheme of a human brain comprising 68 brain regions (Anatomical Desikan-Killiany Atlas (Desikan et al., 2006)), giving rise to a network of interconnected mean fields. The implementation of the model was realized in the TVB platform.

2.5 The case of general anesthesia

To provide an example of the use of our framework, we consider the case of general anesthesia. General anesthesia directly impact various cortical receptors (Moody et al., 2021), primarily $GABA_A$ and NMDA receptors (Hemmings et al., 2005), influencing the activity of both inhibitory and excitatory synapses. GABAergic anesthetics (propofol, volatile anesthetics) act as a positive allosteric modulator of the $GABA_A$ receptor, potentiating GABA-mediated inhibition (Vanlersberghe and Camu, 2008). They increase the opening duration of the chloride channel, increasing the Cl^- conductance and prolonging the duration of inhibitory post-synaptic potentials (IPSPs) (Forman SA, 2008; Franks, 2008; Nguyen and Postnova, 2021). Conversely, NMDA receptor antagonism (ketamine, xenon), leads to a shortening of the kinetics of the conductance of AMPA/NMDA excitatory synaptic input (Orser et al., 1997; Dilger, 2002; Mion and Villevieille, 2013). Thus, the main effect of these anesthetics on cortical synapses can be described as an increase in the time constant of inhibitory synaptic decay (τ_i in Eq. 4) to account for the IPSP prolongation caused by GABAergic anesthetics, while a decrease in the excitatory synaptic decay time (τ_e in Eq. 4) corresponds to NMDA-antagonism.

Despite these seemingly small changes at the cellular scale, at the meso- and macro-scale, general anesthesia is characterized by a qualitative transition in the global neuronal dynamics. While the awake state is characterized by asynchronous and irregular dynamics, one of the hallmarks of general-anesthesia states is the appearance of high-amplitude, low-frequency (<4 Hz) oscillations in neuronal activity, correlated with synchronized transitions of neocortical neurons between hyperpolarized and depolarized states (Steriade et al., 1993a; Contreras and Steriade, 1995; Steriade et al., 2001). These transitions manifest as alternating periods of intense neuronal firing (UP states) and periods of quiescence (DOWN states).

This type of activity is also found in NREM sleep (Steriade et al., 1993a,b; Timofeev et al., 2001; McCormick, 1993), where the low levels of acetylcholine lead to the activation of leak K^+ channels and the increase of neuronal spike-frequency adaptation (described by the parameter b in the AdEx model). as shown in previous modelling works (Di Volo et al., 2019; Goldman et al., 2022). Variations in the level of acetylcholine are also observed for anesthetics such as propofol (Huang et al., 2024), but it does not constitute its main pharmacological effect. Thus, how this global transition in neuronal dynamics occurs during anesthesia and how it is related to the molecular changes described before is still an open question. Furthermore, how the molecular changes occurring during NREM sleep lead to the same kind of global transition as in anesthesia is yet not well understood. We show in the next sections how our framework can capture this transition at multiple scales and how the mechanisms and interplay between the different molecular mechanisms acting during anesthesia and sleep can be understood using our framework.

2.6 Emergence of UP-and-DOWN states in anesthesia and NREM sleep

In this section, we present an analysis of the mechanisms behind the emergence of UP-and-Down states in general anesthesia and its relation to slow waves in NREM sleep. Previous modeling work has shown that the emergence of Up-and-Down states during NREM sleep can be associated with the periodic destabilization of a self-sustained state due to the action of a slow adaptation current (Di Volo et al., 2019; Cakan et al., 2022). For a large enough strength of

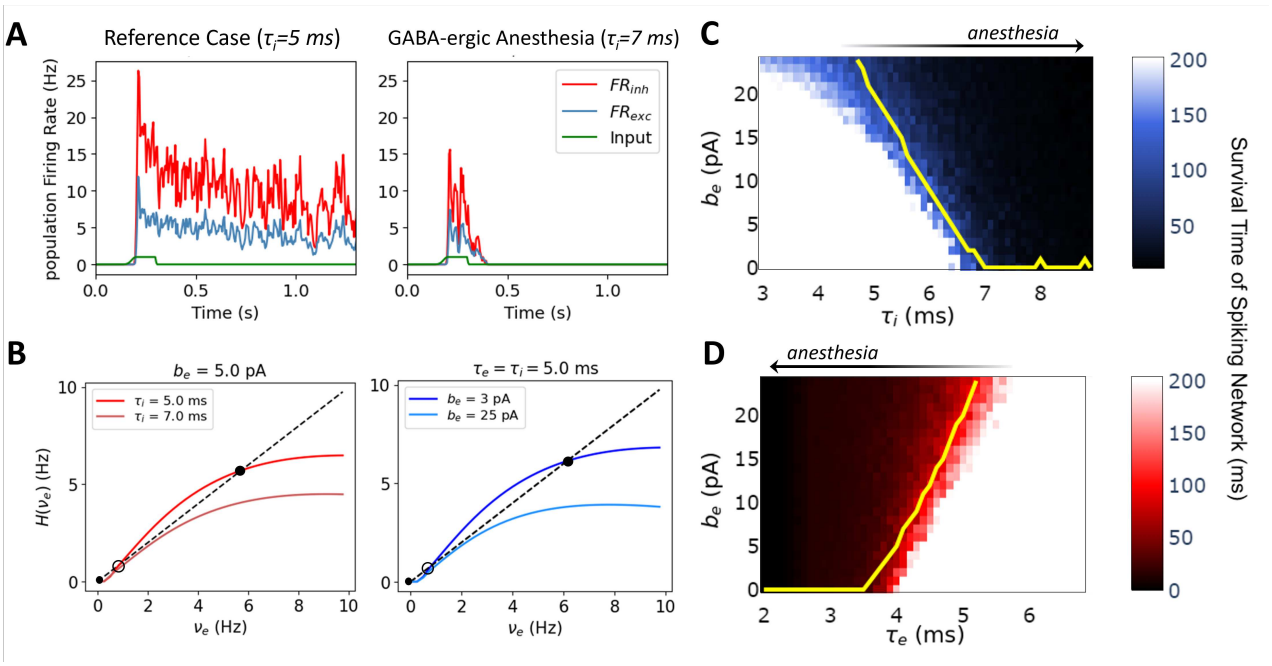

Figure 3: Emergence of UP-and-DOWN states in simulated anesthesia and NREM sleep: effects of molecular mechanisms at the mesoscale level. **A**. Loss of self-sustain activity states in the spiking network as a consequence of changes in GABA-ergic receptors under general anesthesia (for GABA-targeting case). An external input (depicted in green) stimulates the network, resulting in a high firing rate. In the case of higher τ_i (general anesthesia state, right panel), the system's activity is suppressed after the end of the stimulus, in contrast to the reference case (lower values of τ_i, left panel) where a self-sustained state is obtained. **B**. Graphical solution of the first-order mean-field equations for stationary points, predicting the destabilization of self-sustain states and the emergence of the UP-and-DOWN states in NREM sleep and general anesthesia. The function $H(\nu_e)$ corresponds to the Transfer Function of pyramidal cells evaluated at the corresponding stationary solution for FS cells and the black dashed line corresponds to the bisector (see methods and ref. Di Volo et al. (2019) for details). Stable fixed points are marked with filled circles and unstable ones with empty circles. The destabilization of the high-rate (self-sustained) stationary point is promoted by the increase of the synaptic decay time τ_i (left panel) and due to the increase of spike-triggered adaptation (b_e, right panel). **C,D**. Decay time of the spiking network activity after the end of an external stimulus (survival time of the spiking network) as a function of synaptic decay times and spike-triggered adaptation parameters. For each value of $\tau_{e,i}$ a critical value of b_e exists (b_{crit}) for which the sustained network activity is suppressed. For each combination of b_e and $\tau_{e,i}$, 20 different trials with different noise realizations were conducted. The final survival time is reported as the average across these trials. The yellow lines describe the superimposed mean-field predictions of b_{crit} (see text for details), demonstrating the capacity of the mean-field to explain the transition in dynamics at the mesoscale level.

the spike-triggered adaptation current, determined by the parameter b in Eq. 2, a transition from an asynchronous irregular state to a UP-and-DOWN state is obtained. Here we extend this analysis and show by combining the mean-field and network simulations (mesoscale level) that this destabilization is promoted by the variations in the synaptic decay times (induced by general anesthesia) and, furthermore, that the transition to UP-and-DOWN states in general anesthesia can be explained as a result of an interplay between the values of synaptic decay $\tau_{e,i}$ and the spike frequency adaptation of excitatory cells b_e (related to neuromodulatory currents that affect neuronal excitability).

We start by examining the self-sustainability of the AdEx spiking networks as a function of the parameters b_e and $\tau_{e,i}$ (τ_e or τ_i). To investigate the existence of self-sustained states in the network we apply a short initial stimulus of 1Hz of amplitude and a total duration of 120ms. In the cases where the system exhibits a self-sustained state, the network maintains its activity after the end of the stimulus, otherwise, it decays fast to zero (Fig. 3.A). An extensive scan along these variables was performed and the survival time of the activity after the end of the stimulus was calculated. Two parameter scans were performed, one for varied b_e and τ_i (Fig.3.C) and one for varied b_e and τ_e (Fig.3.D).

As seen in the Fig. 3.C,D, for both cases there is a region of the parameter space where the network sustains its activity beyond the removal of the external input (white area), followed by a transition zone where the survival time decreases progressively until it reaches an area where the network no longer

is able to sustain its activity (in black). It becomes evident that the loss of the stable high state of the system depends on the interaction between the synaptic parameters $\tau_{e,i}$ and the adaptation b_e.

To better understand the interplay of the different parameters (b_e, τ_e and τ_i), and the mechanism behind the emergence of UP-and-DOWN states, we turn now to the mean-field model. An analysis of the stable solutions of the system is performed. We show in Fig. 3.B a graphical solution for the calculation of the fixed points of the excitatory firing rate (ν_e). We continue with the exploration of the bistability of the system as predicted by the AdEX mean-field, seeking to identify the critical value of b_e (defined as the value of b_e where the high-rate fixed point of the system is lost). To obtain this solution a scan is performed for different values of the ν_e and the corresponding function $H(\nu_e) = F_e(\nu_e, \bar{\nu}_i)$ is calculated, where $\bar{\nu}_i = F_i(\nu_e, \bar{\nu}_i)$ (first-order mean-field $\dot{\nu}_i = 0$). The fixed point of the system is obtained when $H(\nu_e) = \nu_e$, therefore the graphical solution corresponds to the intersection point of the plotted function H and the bisector.

To explore the effects of the parameters b_e and $\tau_{e,i}$ on the stable points of the system we fix the value of $\tau_{e,i}$ and we progressively increase the value of b_e starting from zero. As observed in the plot, as the magnitude of b_e increases there is a critical value (b_{crit}) for which the system transitions from two fixed-points to only one fixed point located at zero. However, this critical value depends on the respective value of $\tau_{e,i}$: variations in $\tau_{e,i}$ modify the global excitability of the system, facilitating or hindering the suppression of the self-sustained state due to b_e (or even completely eliminating the self-sustained state independently of the adaptation). We present the predicted b_{crit} as a function of τ_i and τ_e in Fig. 3.C and D respectively superimposed to the predictions of the survival time of the spiking networks (yellow lines in Fig. 3.C and D). As expected, the b_{crit} is lower for higher values of τ_i, with the opposite behavior observed in the case of τ_e. A good agreement is observed between the predictions of the mean-field and the spiking network, with the mean-field predictions delimiting the area after which the values of the survival time are uniformly zero. Notice that for spiking-networks the intrinsic fluctuations due to limit-size effects induce a gradual transition in the survival time which is not captured by the mean-field model.

In Fig.4 we demonstrate the transition of states (from asynchronous irregular to UP-and-DOWN dynamics) at the different scales obtained within our framework. In conclusion, in this section we have shown how we can capture the emergence of UP-and-DOWN states in general anesthesia at the different scales by modeling molecular changes, and how our framework (network-simulations in combination with the mean-field formalism) can be used to explain the mechanisms behind the transition between states.

2.7 Emergent whole-brain effects of microscopic changes

We advance to the next stage by using our framework to assess emergent changes at the whole-brain scale. We analyze changes occurring at the functional and behavioral level between the awake and anesthetized states and compare our model with experimental data of both resting state and stimulus-response paradigms.

State-dependent Resting State Dynamics: Correlation between Structural and Functional Connectivity An increase of similarity between the structural and functional connectivity (SC and FC respectively) during anesthesia has been reported in experimental data (Barttfeld et al., 2015) and is considered one of the neural signatures of unconsciousness. This property can only be studied using a whole brain model capable of describing the activity of distinct brain regions (functional component) while incorporating tractographic information on their connections (structural component). Our approach offers the possibility of studying such emergent properties while allowing the mapping of macroscale observations to microscopic factors.

We used resting state fMRI experimental data from humans in the awake state and under propofol anesthesia to reveal differences in SC-FC correlation between the two states. We followed the same analysis on simulated data, by generating BOLD signals based on the nodes' firing rates of the whole-brain model, and compared the experimental results to the model's predictions.

Figure 5 presents the Pearson correlation between functional and structural connectivity for both experimental and simulated data in humans (Fig. 5). Resting-state fMRI data from 26 individuals during wakefulness and under propofol anesthesia were analysed. From these signals, the FC matrix was computed for each participant and correlated with a reference SC. The Pearson correlation between SC and FC was significantly higher during anesthesia compared to wakefulness.

We used our framework to simulate wakefulness and propofol anesthesia, modeled as $GABA_A$ receptor potentiation. The firing rates were convolved with a canonical hemodynamic response function to generate simulated BOLD signals (done with the BOLD-monitor integrated in the TVB platform). The

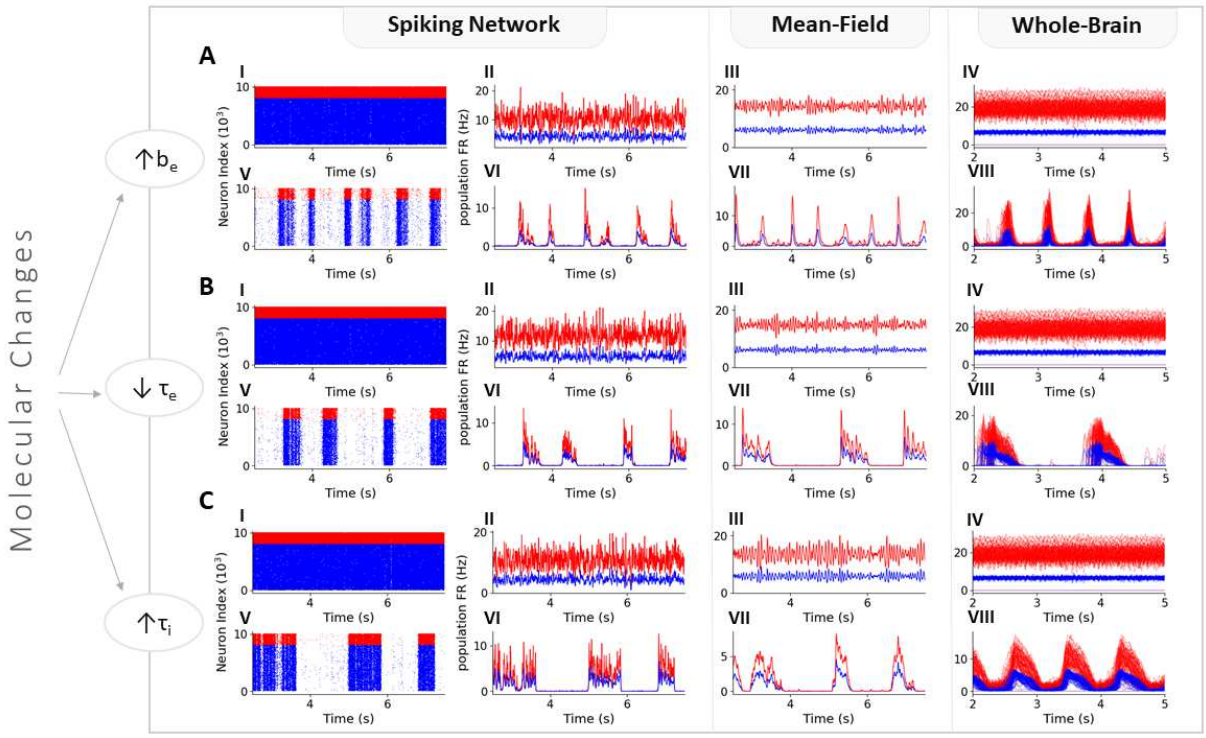

Figure 4: Transition from asynchronous irregular to UP-and-DOWN dynamics across the different components and scales of our framework: spiking network (single-cell models), mean-field and whole-brain model. Raster plots (A-$C_{I,\,V}$) and average firing rate (A-$C_{II,\,VI}$) of the excitatory (blue) and inhibitory (red) AdEx neurons in the spiking network, with the corresponding mean-field dynamics (A-$C_{III,\,VII}$) and whole-brain model (A-$C_{IV,\,VIII}$). For each of the A,B,C cases, a different microscopic parameter (b_e, τ_e, τ_i respectively) is changed to produce synchronous dynamics. **A**. Increased spike-frequency adaptation current b_e simulating NREM sleep, b_e =5pA in AI and 120pA in UD, $\tau_e = \tau_i$ =5ms in both conditions, ν_{drive} =0.4Hz. In the whole-brain model, ν_{drive} =0.315Hz for both AI and slow-wave state. **B**. Decreased excitatory synaptic decay, τ_e, mimicking the action of NMDA-blockers(τ_e =5ms in AI and 3.75ms in UD, b_e =20pA in both conditions,ν_{drive} =0.55Hz). In the whole-brain model, b_e = 5pA and τ_e =5ms in AI, b_e = 30pA and τ_e = 3.75ms in synchronous state and ν_{drive} =0.315Hz in both states **C**. Increased inhibitory synaptic decay, τ_i, mimicking the action of GABA-ergic agents(τ_i =5ms in AI and 7.0ms in UD, b_e =10pA in both conditions,ν_{drive} =0.345Hz). In the whole-brain model, b_e = 5pA and τ_e =5ms in AI, b_e = 30pA and τ_i = 7ms in synchronous state and ν_{drive} =0.315Hz in both states

calculated FC matrices were correlated with the same reference SC used for the experimental data. Different seeds (n=16) for noise realizations were implemented to address inter-subject variability. The whole-brain model successfully captures the same transition between the simulated data for wakefulness and propofol anesthesia. Despite a larger increase of SC-FC correlation in the case of simulated general anesthesia, the model qualitatively reproduced their relative difference.

State-dependent Evoked Dynamics: Perturbational Complexity Index Conscious and unconscious states exhibit distinct brain responses to external stimuli, with a longer and more widespread propagation of brain activity across the brain during conscious states (Massimini et al., 2005). This state-dependent response is quantified and clinically study with the Perturbational Complexity Index (PCI), which can distinguish different states of consciousness by quantifying the complexity of EEG responses after a TMS-induced direct cortical perturbation (Casali et al., 2013; Massimini et al., 2005). Conscious individuals typically demonstrate higher PCI values compared to those during NREM sleep or under general anesthesia (Fig. 6.B).

Here, we adopt a similar approach to explore how perturbations propagate through a network of mean-fields in the whole-brain model. Goldman et al. (2022) demonstrated that simulated NREM sleep, modeled with increased spike-triggered adaptation in the whole-brain model, was characterized by lower PCI values compared to wake-like states. In this study, we extend this analysis to simulated anesthesia, comparing it with both simulated NREM sleep and wakefulness.

The perturbation was introduced in the whole brain model as a square wave of amplitude 1 Hz and a duration of 50 ms, applied to the excitatory population of the caudal part of the right middle

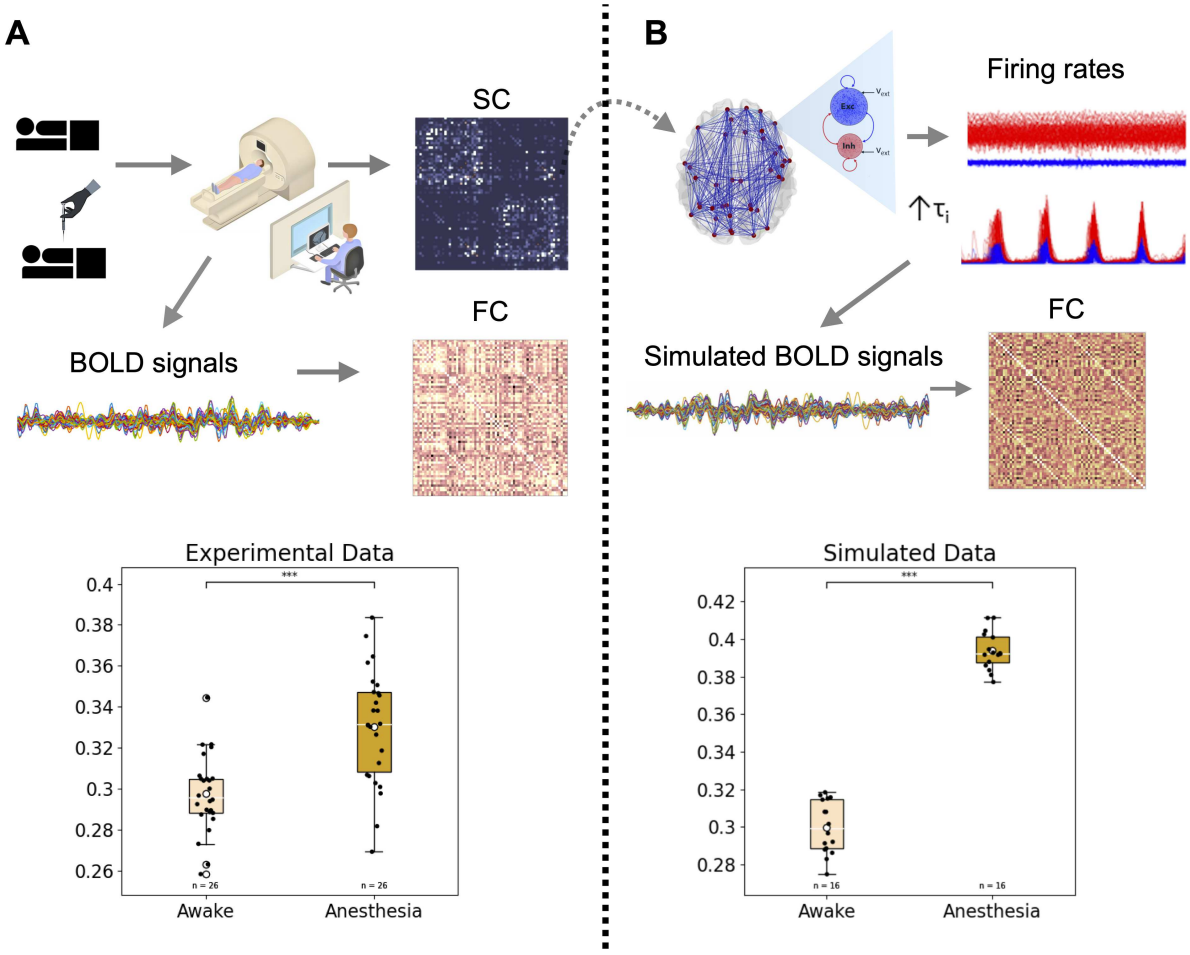

Figure 5: Structure-function correlation is increased under anesthesia. A) Resting-state fMRI BOLD signals from 26 individuals were obtained during wakefulness and under propofol anesthesia and were used to compute the functional connectivity matrix (FC) for each participant. We observe a significantly higher correlation between the structural connecivity (SC) and FC for the case of propofol anesthesia. **B)** The whole-brain model was used to simulate BOLD signals for the wake-like ($b_e = 30$pA, $\tau_i = 5$ms) and propofol anesthesia ($b_e = 30$pA, $\tau_i = 7$ms) conditions (16 different noise realizations). An increased correlation between SC-FC was observed, similar to the experimental data. *** $p < 0.001$ from Mann-Whitney U test. Middle white line: median; white dot: mean; box limits: first quartile (Q1) and third quartile (Q3) of the data; whiskers: 1.5x inter-quartile range (IQR); Black dots show the SC-FC correlation per subject or noise realization.

frontal gyrus. In Figure 6.C, we provide a graphical representation of the spatiotemporal aspects of the signal propagation. The plot illustrates the time each region exceeded the significance activation threshold. The results indicate that, in the case of simulated wakefulness, the signal propagated extensively across the cortex, with regions becoming activated late (depicted in blue). In contrast, during simulated unconscious states, only a few regions surpassed the significance threshold, and these activations occurred primarily shortly after the initial stimulus.

In Figure 6.A we show the PCI values calculated across 60 trials (using different noise seeds) per condition. We observe that simulated wakefulness is associated with higher PCI values compared to all other unconscious states, namely NREM sleep , NMDA-blocker anesthesia , and GABA$_A$-potentiators anesthesia.

While PCI values in wake-like condition were significantly higher than the unconscious states ($p<0.001$), a high degree of overlap was found among the unconscious, with no significant differences detected between the different simulated mechanisms of general anesthesia, although both were lower than NREM sleep.

3 Discussion

We present a new computational framework to simulate emergent dynamics and brain states arising from changes at the molecular scale. Our approach utilizes a multiscale model that begins at the cellular level and follows a bottom-up methodology to reach the whole-brain scale. A key aspect of our approach is

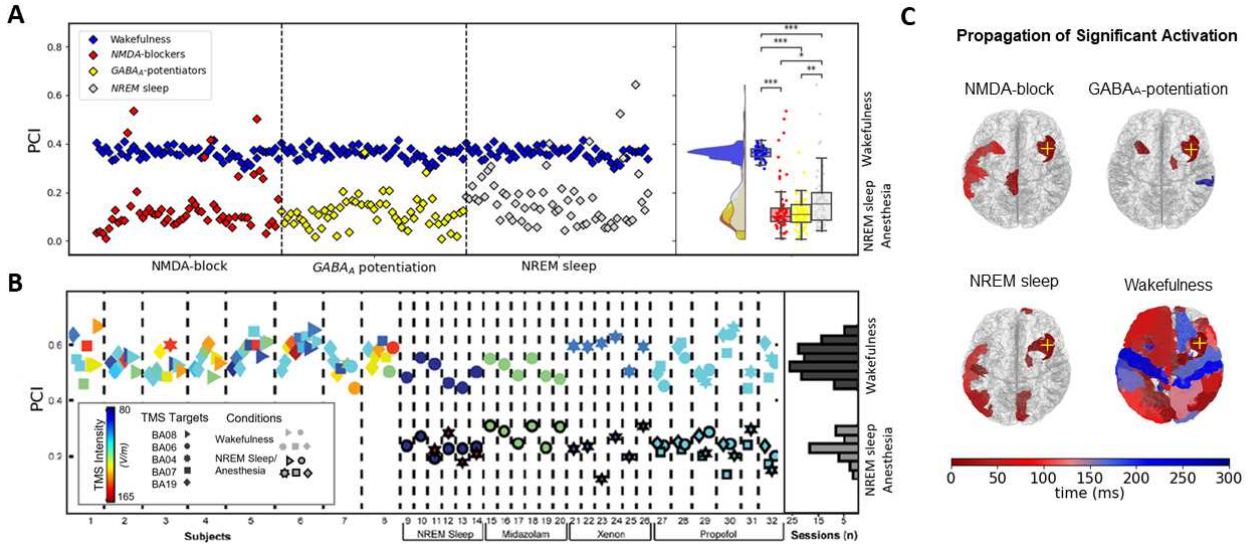

Figure 6: Reduced responsiveness to external stimulus in unconscious states (simulated (A,C) and empirical (B)). **A**. Perturbational Complexity Index (PCI) values for the cases of simulated wakefulness ($b_e = 5$pA, $\tau_e = \tau_i = 5.0$ms), anesthesia (NMDA-blockers ($\tau_e = 3.75$ ms, $b_e = 30$pA), GABA$_A$ receptors potentiation($\tau_i = 8.0$ms, $b_e = 30$pA)) and NREM sleep ($b_e = 120$pA, $\tau_e = \tau_i = 5.0$ms). Each point corresponds to a different noise realization of the model (in total 60 trials per condition). The horizontal axis corresponds to the noise seed used for the specific simulation, so that each pair of wakefulness-unconscious state is plotted on the same vertical axis; in the most right panel the raw jittered data are collapsed to the same vertical axis for each condition; the limits of boxplots include the first quartile (Q1) and third quartile (Q3) of the data, the middle horizontal lines correspond to the median values; the half violin plots describe the probability density function; Post-hoc Conover tests were performed to assess statistical significance of the difference in PCI values between the conditions, and the Holm-Bonferroni method was used to control for multiple comparisons (*: p<0.05, **: p<0.01, ***: p<0.001). **B**. Modified from Casali et al. (2013): Empirical PCI data from human subjects showing a clear distinction between awake and unconscious individuals. **C**. Spatiotemporal dynamics of applied perturbation propagation. Following stimulation, applied at $t = 0$ms, all regions were evaluated to determine if they surpassed the significance threshold, as defined by the PCI calculation algorithm. The time each region exceeded this threshold is color-coded showcasing a widespread and temporally sustained signal propagation during wakefulness. The yellow cross denotes the stimulated region, corresponding to the caudal part of the right middle frontal gyrus.

the use of biophysically grounded mean-field models to bridge the gap between the micro and macro scales, enhancing computational efficiency by reducing the system dimensionality. Additionally, the bottom-up mean-field models developed and implemented in our framework incorporate all cellular parameters, which allows us to capture the emergence of different brain states and brain dynamics generated by changes at the cellular level and to understand the mechanisms behind these transitions.

To demonstrate the application of our framework, we study the effects of common anesthetics at the whole brain level. Our approach successfully captures the different mechanisms leading to changes in brain states (between conscious and unconscious states) in sleep and anesthesia, highlighting the generality of our approach. Furthermore, we illustrate how our model can explain the transition between dynamical states and the complex interplay between different molecular parameters and mechanisms underlying these transitions. These characteristics make our model suitable for modeling anesthetics in biologically realistic scenarios. Previous studies have used the mean-field approach to model the macroscopic effects of general anesthesia on the brain at the macroscale (Luppi et al., 2022), and the action of psychedelic compounds on the brain (Deco et al., 2018; Herzog et al., 2023), however, without explicitly considering the biophysical details of the molecular action.

Our model holds great potential for clinical applications, where it could be used to predict particular activity patterns as a function of anesthesia depth, such as the level of diminished responsiveness, which may be interesting to explore for intermediate anesthesia levels.

To yield more precise models providing a quantitative comparison with experimental data, it is crucial to accurately reconstruct brain signals (such as EEG, MEG or MRI) from mean-field models, as proposed recently (Tesler et al., 2022, 2023b), which will allow a more direct comparison with experiments. In addition, it is of great importance to include regional variations, by designing specific models for each

brain area, as done recently for non-cortical structures (Overwiening et al., 2023; Tesler et al., 2023a, 2024). We hope that this level of biophysical accuracy will bring the model closer to a quantitative account of the fine details of the experimental data, and be more precise and specific to investigate their underlying mechanisms.

4 Methods

We describe here the details of the multiscale whole-brain model: we start at the cellular level with the introduction of the single-cell model, then we move to the mesoscale with the construction of a neuronal network and the formulation of a mean-field model describing the mesoscopic network activity, and we end at the whole brain scale using a large network of mean-fields interconnected using the anatomical connectome, step realized through the integration in The Virtual Brain simulator (Fig. 1).

4.1 Single cell model

The dynamics of the single neuron are based on the adaptive exponential integrate-and-fire model (AdEx) and are described by the following equations (Brette and Gerstner, 2005):

$$c_m \frac{dv}{dt} = g_L(E_L - v) + g_L \Delta e^{\frac{v - v_{thr}}{\Delta}} - w + I_{syn} \quad (1)$$

$$\frac{dw}{dt} = -\frac{1}{\tau_w}(a(v - E_L) - w) + b \sum_{t_{sp}} \delta(t - t_{sp}) \quad (2)$$

where v is the membrane potential, $c_m = 200$ pF is the membrane capacitance and $g_L = 10$ nS is the leak conductance. The leakage reversal potential E_L equals -65 mV and -64 mV for inhibitory and excitatory cells respectively. The exponential term Δ has a different strength for the two types of cells (2 mv for excitatory and 0.5 mV for inhibitory cells). W and I_{syn} describe the adaptation and synaptic current respectively. Parameter a describes the sub-threshold adaptation and b describes the spike-triggered adaptation. A spike is generated when the membrane potential exceeds a voltage threshold $v_{thr} = -50$ mV at time t_{sp}. The neuron's membrane potential is subsequently reset to a resting voltage $v_{rest} = -65$ mV and fixed to that value for a refractory period $T_{refr} = 5$ ms. The Dirac δ-function indicates that whenever a neuron fires at time $t_{sp}(k)$, the adaptation current W is incremented by an amount b. Based on physiological characteristics (Destexhe, 2009), the inhibitory cells are modelled as fast-spiking (FS) neurons, exhibiting no adaptation ($a_i = b_i = 0$). Excitatory neurons are modelled as regular spiking (RS) cells, characterised by lower excitability due to the presence of spiking frequency adaptation (b_e varies in our simulations, $a_e = 0$nS, and the adaptation time constant $\tau_w = 500$ms).

4.2 Synaptic model

Each neuron k receives a synaptic current I_{syn}, which corresponds to the spiking activity of all presynaptic neurons $j \in \text{pre}(k)$ of neuron k. I_{syn} can be decomposed to the input received from excitatory (E) and inhibitory (I) presynaptic spikes, so as

$$I_{syn} = G^e_{syn}(E_e - v) + G^i_{syn}(E_i - v), \quad (3)$$

where $E_e = 0$mV ($E_i = -80$mV) is the excitatory (inhibitory) reversal potential and G^e_{syn} (G^i_{syn}) the excitatory (inhibitory) synaptic conductance. Synaptic conductances were modeled by a decaying exponential function that sharply increases by a fixed amount Q_x , at each spiking time (t_{sp,j_x}) of a presynaptic neuron j_x:

$$G^x_{syn}(t) = Q_x \sum_{t_{sp,j_x}} \Theta(t - t_{sp,j_x})\, e^{-\frac{t - t_{sp,j_x}}{\tau_x}}, \quad (4)$$

where x corresponds to the population type ($x \in \{e, i\}$), Θ is the Heaviside function, τ_x is the characteristic decay time of synaptic conductances (varied in our simulations), and Q_x is the quantal conductance ($Q_e = 1.5$nS , $Q_i = 5$nS). The sum runs over all the spiking times of excitatory (t_{sp,j_e}) or inhibitory (t_{sp,j_i}) presynaptic neurons j_e or j_i of neuron k.

4.3 Spiking network model

We studied a network of $N = 10^4$ integrate-and-fire neurons that displayed spike-frequency adaptation (Destexhe, 2009; Zerlaut et al., 2018). The neurons were connected over a topologically random network with a probability of connection between two neurons equal to $p = 5\%$. The network was composed of two populations of inhibitory and excitatory neurons, with the inhibitory neurons consisting the 20% of the whole network size.

4.4 Mean-field model

The mean-field used here is based on a bottom-up formalism derived from a Master Equation (El Boustani and Destexhe, 2009), and which has recently been extended to include the effects of adaptation (Zerlaut et al., 2018; Di Volo et al., 2019). The mean-field equations of the system can be written as :

$$
\begin{aligned}
T\frac{\partial \nu_\mu}{\partial t} &= (F_\mu - \nu_\mu) + \frac{1}{2}c_{\lambda\eta}\frac{\partial^2 F_\mu}{\partial \nu_\lambda \partial \nu_\eta} \\
T\frac{\partial c_{\lambda\eta}}{\partial t} &= \delta_{\lambda\eta}\frac{F_\lambda(1/T - F_\eta)}{N_\lambda} + (F_\lambda - \nu_\lambda)(F_\eta - \nu_\eta) \\
&\quad + \frac{\partial F_\lambda}{\partial \nu_\mu}c_{\eta\mu} + \frac{\partial F_\eta}{\partial \nu_\mu}c_{\lambda\mu} - 2c_{\lambda\eta} \qquad (5) \\
\frac{\partial W}{\partial t} &= \frac{1}{\tau_w}(a(\mu_V(\nu_e, \nu_i, W) - E_L) - W) + b\nu_e \ ,
\end{aligned}
$$

where $\mu = \{e, i\}$ is the neural population index (excitatory or inhibitory), ν_μ the mean firing rate of the corresponding population, $c_{\lambda\eta}$ the covariance between populations (λ and η) , W is the mean adaptation, and T is the mean-field characteristic time constant.

The function $F_\mu = F_\mu(\nu_e, \nu_i, W)$ is the transfer function (TF) of a neuron of type μ, i.e. its output firing rate when receiving inhibitory and excitatory inputs with rates ν_e and ν_i, and adaptation level W. The TF can be derived following a semi-analytic approach (Zerlaut et al. (2018)), where the output firing rate of a neuron can be written as a function of its mean subthreshold membrane voltage μ_V, its standard deviation σ_V, and its time correlation decay time τ_V:

$$
F = \nu_{\text{out}} = \frac{1}{2\tau_V} \cdot \text{Erfc}\left(\frac{V_{\text{thr}}^{\text{eff}} - \mu_V}{\sqrt{2}\sigma_V}\right). \qquad (6)
$$

where $(\mu_V, \sigma_V, \tau_V)$ are calculated as a function of the input firing rates (ν_E, ν_I) and the adaptation intensity W following the equations described in Di Volo et al. (2019). The $V_{\text{thr}}^{\text{eff}}$ is the phenomenological spike threshold voltage taken as a second-order polynomial:

$$
\begin{aligned}
V_{\text{eff}}^{\text{thr}}(\mu_V, \sigma_V, \tau_V^N) = P_0 &+ \sum_{x \in \{\mu_V, \sigma_V, \tau_V^N\}} P_x\left(\frac{x - x_0}{\delta x_0}\right) \\
&+ \sum_{x,y \in \{\mu_V, \sigma_V, \tau_V^N\}^2} P_{xy}\left(\frac{x - x_0}{\delta x_0}\right)\left(\frac{y - y_0}{\delta y_0}\right),
\end{aligned}
$$

where $\tau_V^N = \frac{\tau_V G_l}{C_m}$ a non-dimensional quality. The polynomial coefficients of P are determined through a fitting of the TF template to the output firing rate of individual neuron simulations, varying both inhibitory and excitatory inputs. The values for the normalization of the fluctuation regime were set following previous work (Zerlaut et al., 2018; Di Volo et al., 2019): μ_V^0 = -60 mV, $\sigma_V^0 = 0.004$ mV, $(\tau_V^N)^0 = 0.5$, $\delta\mu_V^0 = 0.001$ mV, $\delta\sigma_V^0 = 0.006$ mV and $\delta(\tau_V^N)^0 = 1$. The fit is performed for each of the cell types included in the network, so in our case for fast and regular spiking cells. As shown in Di Volo et al. (2019) the mean-field predictions work even far from the fitting point of the TF, so in all the examples showcased in this work we used previously calculated P, fitted for similar parameterizations of the two types of cells (see Table 1).

Taking the first-order equations by disregarding the covariance term dynamics in Equation 5, the mean-field model for the excitatory and inhibitory populations representing a cortical volume can be written as:

$$
\begin{aligned}
T\frac{d\nu_e}{dt} &= \mathcal{F}_e(\nu_e + \nu_{aff}, \nu_i, W) - \nu_e \\
T\frac{d\nu_i}{dt} &= \mathcal{F}_i(\nu_e + \nu_{aff}, \nu_i, W) - \nu_i \qquad (7) \\
\frac{dW}{dt} &= -\frac{W}{\tau_w} + b\nu_e + \frac{a}{\tau_w}(\mu_V(\nu_e, \nu_i, W) - E_L) \ ,
\end{aligned}
$$

where ν_{aff} denotes an afferent input to both type of populations which we write as:

$$
\nu_{aff}(t) = \nu_{drive} + \sigma\xi(t), \qquad (8)
$$

where $\sigma = 3.5$ and $\xi(t)$ denotes an Ornstein-Uhlenbeck (OU) process, of the form:

$$
d\xi(t) = -\xi(t)\frac{dt}{\tau_{OU}} + dW_t \qquad (9)
$$

where $\tau_{OU} = 5$ms is the timescale of the OU process and dW_t is a Wiener process of amplitude one and zero average. The ν_{drive} received different values in our simulations depending on the microscopic parameter that was selectively changed.

4.5 Networks of mean-field models

To model whole-brain dynamics, a network of mean-fields is defined, where each mean-field describes the activity of a brain region. The interactions between the mean-fields are excitatory while the inhibitory connections are preserved solely on the regional level. Extending on the single mean-field equations (Eq. 7), now the population activity of each region is given by the following equations:

$$
\begin{aligned}
T\frac{d\nu_e(k)}{dt} &= \mathcal{F}_e\left[\nu_e^{input}(k), \nu_i(k), W(k)\right] - \nu_e(k) \\
T\frac{d\nu_i(k)}{dt} &= \mathcal{F}_i\left[\nu_e^{input}(k), \nu_i(k), W(k)\right] - \nu_i(k) \\
\tau_w\frac{dW(k)}{dt} &= -W(k) + b\tau_w\nu_e(k) \\
&\quad + a(\mu_V(\nu_e(k), \nu_i(k), W(k)) - E_L) \ ,
\end{aligned}
$$

where $\nu_e(k)$ ($\nu_i(k)$) describes the firing rates of the excitatory (inhibitory) population of the region (k), $W(k)$ the adaptation of the population, and $\nu_e^{input}(k)$ is the total excitatory synaptic input that the region

Cell Type	P_0	$P_{\mu V}$	$P_{\sigma V}$	$P^V_{\tau N}$	$P^2_{\mu V}$	$P^2_{\sigma V}$	$P^V_{(\tau N})^2$	$P_{\mu V}\sigma V$	$P_{\mu V}\tau N^V$	$P_{\sigma V}\tau N$
RS-cell	-50.17	4.52	-7.94	-2.08	-0.55	3.42	-11.56	1.95	2.74	-10.67
FS-cell	-51.85	6.16	-14.04	1.67	-2.06	3.18	-31.13	6.57	1.72	-45.16

Table 1: Fitting parameters expressed in mV

receives from the rest of the nodes of the network, given by the following equation:

$$\begin{aligned} \nu_e^{input}(k) &= \nu_e(k) + \nu_{aff}(k) \\ &\quad + G \sum_j C_{jk}\, \nu_e(j, t - \|j-k\|/v_c) \end{aligned}$$

where the sum runs over all nodes j connected to node k, and C_{jk} the connection strength between j and k (and is equal to 1 for $j = k$), and the global coupling factor G rescales the connection strength while maintaining the ratio between different values. The term $\|j-k\|$ is the distance between the nodes j and k, while v_c is the speed with which the signal propagates along the axis so that the model accounts for the delay of axonal propagation. Here ν_{aff} is defined as in Eq. 8.

A cortical parcellation of 68 regions was used. The connection strengths and tract lengths between the nodes were provided by the Berlin empirical data processing pipeline, based on human tractography methods (Schirner et al., 2015). We performed the simulations using The Virtual Brain (TVB) platform, leveraging its capabilities to model and simulate large-scale brain networks. TVB offers a useful tool to study the dynamics of the entire brain through the construction of a network defined by its large-scale structural connectivity and the mesoscopic models that describe each node's intrinsic dynamics. This approach allows us to simulate and analyze the complex interactions and emergent properties of the brain's functional dynamics at both the mesoscopic and macroscopic levels.

4.6 Data for Structural-Functional Connectivity Analysis

Simulated BOLD signals The simulated BOLD signal was generated through the dedicated *BOLD* monitor provided by The Virtual Brain software. The simulated mean field time series of the excitatory population (down-sampled at 250Hz) of each node were convolved with the hemodynamic response function (described by the first order Volterra kernel of the Ballon Windekessel model (Friston et al., 2000)). This signal was finally down-sampled to the frequency matching the repetition time (TR) of the fMRI series. For the simulations of the human brain, the human connectome described in 4.5 was used.

Human fMRI Data Details can be found in (Huang et al., 2021).

4.7 Evoked activity and PCI calculation

The Perturbational Complexity Index (PCI), as proposed by Casali et al. (Casali et al., 2013) is the normalized Lempel-Ziv complexity of the evoked spatiotemporal patterns of cortical activation $SS(x,t)$ and was computed to capture the complexity of the signal propagation after a stimulation. Evoked potentials were generated using the whole-brain model, where a square wave stimulus was applied on the firing rates of the excitatory population on a single node (corresponding to the right premotor cortex). 60 different trials, each with a different stimulus onset and noise realization, were performed where the analysis concerns an interval of 300 ms pre- and post-stimulus. The firing rates of the excitatory populations were firstly binarized to extract significant vectors $s(t)$ per trial, and the Lempel-Ziv complexity $LZ(S)$ was calculated for each of these vectors (Goldman et al., 2022). Finally the PCI is expressed as the ratio of $\frac{LZ(S)}{H(S)}$ where $H(S)$ is the spatial source entropy:

$$\begin{aligned} H(S) = &-p(S=0)\log_2(p(S=0)) \\ &- p(S=1)\log_2(p(S=1)) \end{aligned} \tag{10}$$

4.8 Data availability

Data used in this article can be found in `https://zenodo.org/records/6955280`

4.9 Code availability

The code source of all simulations shown in this article is available online in `https://github.com/mariasacha/paper_pipeline_hub/`.

Acknowledgments

Research supported by the CNRS and the European Union (Human Brain Project H2020-785907, H2020-945539, Virtual Brain Twin project 101137289).

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